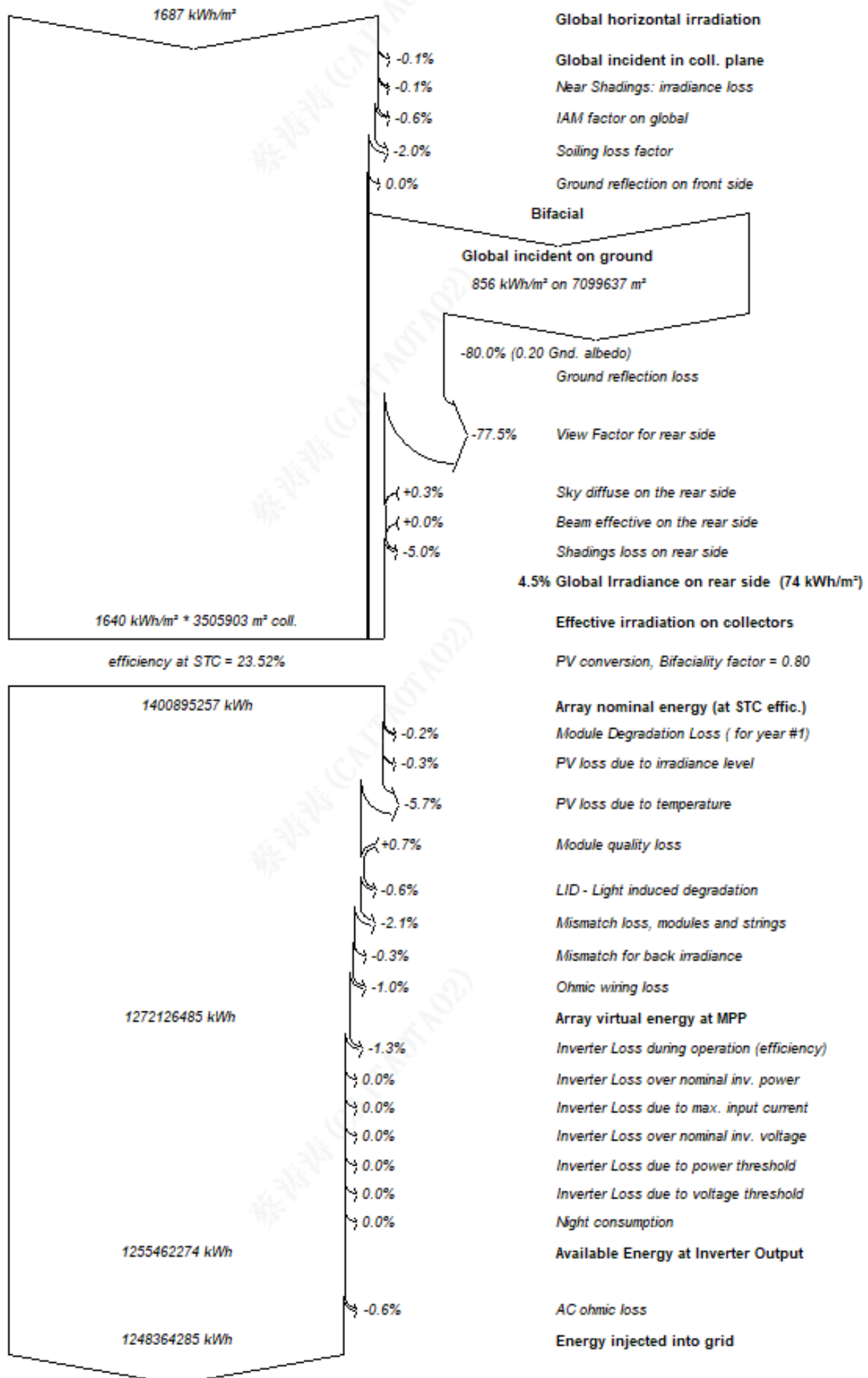
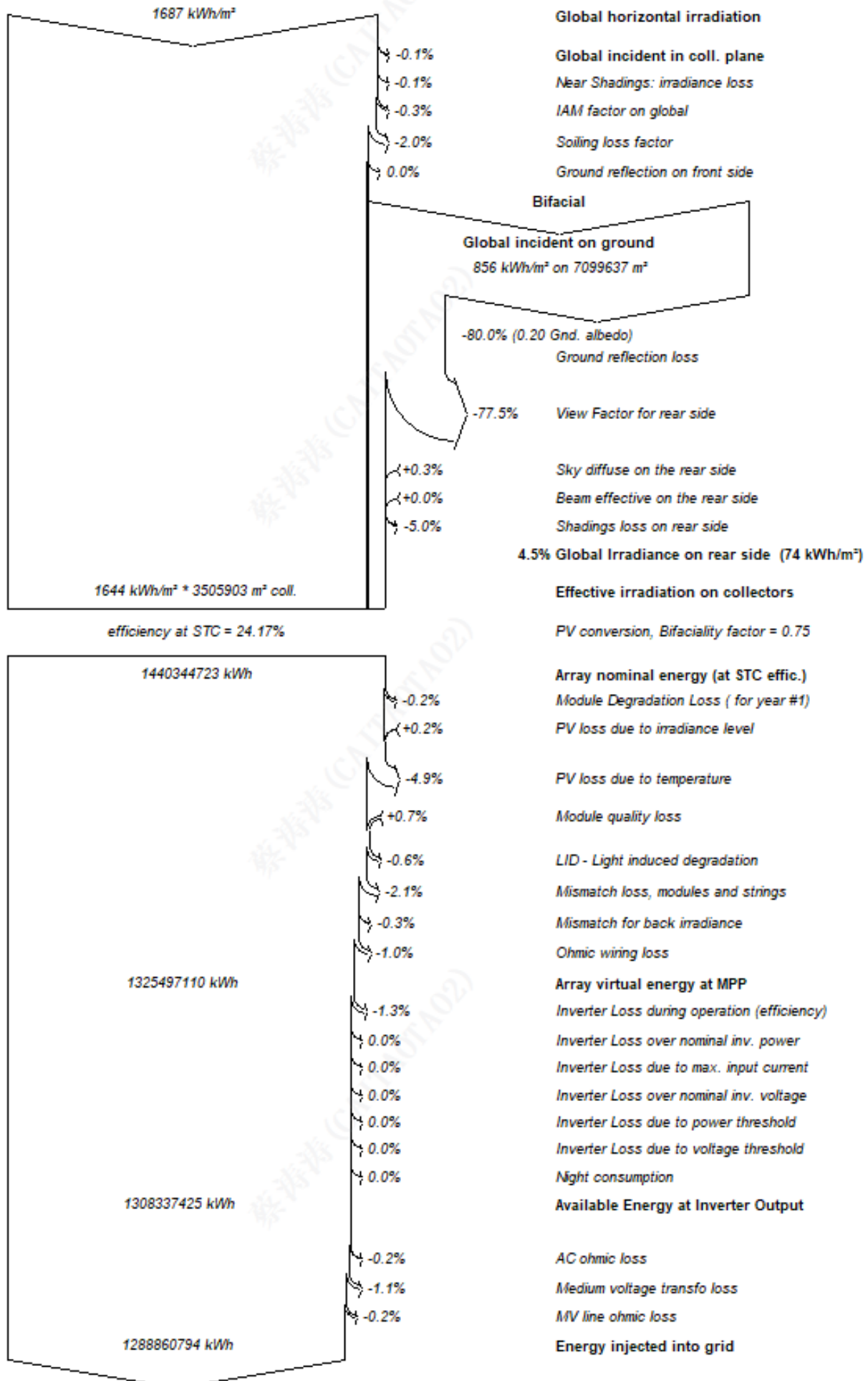


Loss diagram for "TOPcon 730" - year



Loss diagram for "BC750 (0807)" - year



已知 TOPcon 组件温度系数-0.29%/度，BC 组件温度系数-0.26%，两种组件在相同辐照下，温度系数导致的发电量损失如何定量计算？

一、基础计算公式

功率温度损耗通用公式：

发电量损失功率 = 标准功率 × 温度系数绝对值 × 工作温升

发电量损失比例 = 温度系数绝对值 × 工作温升

参数说明：

1. 温度系数：按组件 Datasheet 标定数据，TOPcon = -0.29%/°C，BC = -0.26%/°C
2. 标准测试温度 STC：25°C
3. 工作温升 ΔT = 组件实际工作温度 - 25°C

二、单种组件发电量损失计算

1. TOPcon 组件发电损失率 = $0.29\% \times \Delta T$
2. BC 组件发电损失率 = $0.26\% \times \Delta T$

三、两者发电量差值（核心对比）

同温度环境下，BC 相对 TOPcon 减少的发电损失比例：

差值损失率 = $(0.29\% - 0.26\%) \times \Delta T = 0.03\% \times \Delta T$ ①

当支架形式为倾斜固定支架时，可作以下修正：

根据西安总部实测（组件安装于平屋面上安置的倾斜固定支架，通风良好；不是紧贴屋面，如直接固定在彩钢瓦屋面），BC 组件相较于 TOPcon 组件，工作温度有-0.5°C温差。因此，两种组件的发电损失比率差值为：

差值损失率' = $0.29\% \times \Delta T - 0.26\% \times (\Delta T - 0.5)$ ，差异大于①式

四、实操举例计算

举例：假设组件工作温度 50°C（具体计算，见“五、组件工作温度计算”）

工作温升 $\Delta T = 50 - 25 = 25^\circ\text{C}$

1. TOPcon 总发电损失： $0.29\% \times 25 = 7.25\%$
2. BC 总发电损失： $0.26\% \times (25 - 0.5) = 6.37\%$
3. BC 比 TOPcon 多发电量比例： $7.25\% - 6.37\% = 0.88\%$ (与本文开头损失图中的比例接近)

五、组件工作温度计算

公式来源：PVsyst 技术文档—— $T_{\text{cell}} = T_{\text{amb}} + 1/U \times (\text{Alpha} \times G_{\text{inc}} \times (1 - \text{Effic}))$ ，如下图帮助文档所示。

T_{amb} ——环境温度，可取一年中发电时段（如 6:30~17:30）内环境温度平均值；

U ——热损失系数，空气通常取 29，屋顶取 27；

Alpha ——PVsyst 取 0.9；

G_{inc} ——组件表面总辐照量，对应发电时段的辐照平均值；

Effic ——组件功率；

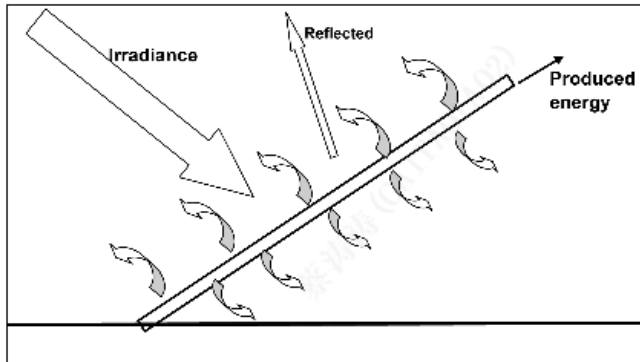
通过上面数据，可计算出组件年平均工作温度，进而算得工作温升 ΔT 。

Thermal Model

The objective is to evaluate the array (or cell) temperature during the simulation. The cell temperature is a basic input parameter of the one-diode model. This is evaluated by an energy balance, accounting for all incoming and outgoing energy fluxes in the array.

At the thermal equilibrium, the energy fluxes should be compensated by the array cooling thermal loss, which is mainly convective:

$$G_{inc} \cdot \text{Alpha} \cdot (1 - \text{Effic}) = U \cdot (T_{cell} - T_{amb})$$



where:

- G_{inc} is the incoming **irradiance** on the module or PV array. PVsyst uses G_{lobInc} for simplicity, although the relevant value should be the effective irradiance.
- Alpha is the **absorption coefficient** of solar irradiance, i.e. $(1 - \text{reflection})$. The usual value chosen for the Absorption coefficient Alpha is 0.9. It is ever choice of Alpha.
- Effic is the **PV efficiency** (related to the module area), i.e. the electrical energy removed from the module. When possible, the PV efficiency is calculated.
- T_{amb} is the ambient temperature, according to the meteo data,
- U is the "**heat loss factor**", expressed in $[W/m^2 \cdot K]$. This is a heat transfer coefficient, determining the heat flux as proportional to the temperature difference.

The left side of this equation describes the different energy fluxes specific to a PV array, the right side defines the necessary heat transfer for ensuring the thermal equilibrium. From this equation, we can easily extract the cell temperature:

$$T_{cell} = T_{amb} + 1 / U \cdot (\text{Alpha} \cdot G_{inc} \cdot (1 - \text{Effic}))$$